

Recap

Total Probability Theorem

$$\Rightarrow P(B) = \sum_{i=1}^n P(B \cap A_i)$$

Bayes rule

$$\Rightarrow P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_{j=1}^n P(B|A_j)P(A_j)}$$

Independence

Conditional Independence

$$\rightarrow P(A \cap B) = P(A)P(B)$$

$$\rightarrow P(A \cap B|C) = P(A|C)P(B|C)$$

Independence $\Rightarrow$ conditional Independence

A: Head in first toss

B: Head in second toss

C: Outcomes of two tosses are different

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A-LH-1

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$$P(A \cap B|C) = 0$$

$$P(A|C) = P(B|C) = \frac{1}{2}$$

Conditional Independence

$\Rightarrow$ pairwise independence

Consider an experiment where you pick a coin from two available coins with equal probability

first coin sees head w.p. 0.9

second " " w.p. 0.2

and you toss with the chosen coin twice & the tosses are independent of each other

C : event that first coin is picked.

A : event that first toss outcome is head

B : " " second " " is head

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addition

$$P(C) = P(C^c) = 0.5$$

$$P(A|C) = P(B|C) = 0.9$$

$$\Rightarrow P(A^c|C) = P(B^c|C) = 0.1$$

$$P(A|C^c) = P(B|C^c) = 0.2$$

$$\Rightarrow P(A^c|C^c) = P(B^c|C^c) = 0.8$$

By the defn of experiment

$$\left. \begin{aligned} P(A \cap B|C) &= P(A|C) P(B|C) \\ P(A \cap B|C^c) &= P(A|C^c) P(B|C^c) \end{aligned} \right\} \text{given}$$

$$\begin{aligned}
 P(A) &= P(A|C)P(C) + P(A|C^c)P(C^c) \\
 &= 0.1 \times 0.5 + 0.2 \times 0.5 \\
 &= 0.55
 \end{aligned}$$



$$\begin{aligned}
 P(B) &= 0.55 \text{ (check!)} \\
 &= P(B|C)P(C) + P(B|C^c)P(C^c)
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow P(A)P(B) &= (0.55)^2 \\
 &\neq P(A \cap B) \\
 &= 0.425
 \end{aligned}$$

$$\begin{aligned}
 P(A \cap B) &= P(A \cap B \cap C^c) + P(A \cap B \cap C) \\
 &= P(A \cap B | C^c)P(C^c) + P(A \cap B | C)P(C) \\
 &= P(A|C^c)P(B|C^c)P(C^c) + P(A|C)P(B|C)P(C) \\
 &= (0.2)^2 \frac{1}{2} + (0.1)^2 \frac{1}{2} = \frac{0.85}{2} = 0.425
 \end{aligned}$$

Independence of multiple events

Events $A_1, A_2, \dots, A_n$ are said to be independent if

$$P\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} P(A_i)$$

for any $S \subseteq \{1, 2, \dots, n\}$

For $n=3$ for A_1, A_2, A_3 to be independent

$$P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2)P(A_3)$$

$$\text{Pairwise independence} \left\{ \begin{array}{l} P(A_1 \cap A_2) = P(A_1)P(A_2) \\ P(A_2 \cap A_3) = P(A_2)P(A_3) \\ P(A_1 \cap A_3) = P(A_1)P(A_3) \end{array} \right.$$

$$P(A_1) = \frac{1}{2} = P(A_2) = P(A_3)$$

Example



$$A_2 \cap A_3 = \{TH\}$$

$$A_1 \cap A_3 = \{HT\}$$

$$P(A_2 \cap A_3) = \frac{1}{4} = P(A_2)P(A_3)$$

$$P(A_1 \cap A_3) = \frac{1}{4} = P(A_1)P(A_3)$$

$\Rightarrow A_1, A_2, A_3$ are pairwise independent

$$P(A_1 \cap A_2 \cap A_3) = 0 \neq P(A_1)P(A_2)P(A_3)$$

Example 1 $A : \{\text{first die } 1, 2 \text{ or } 3\}$
 $B : \{\text{first die } 3, 4, 5\}$
 $C : \{\text{sum of two dice is } 9\}$

$$A = \{1, 2, 3\} \times \{1, 2, 3, 4, 5, 6\}$$

$$B = \{3, 4, 5\} \times \{1, 2, 3, 4, 5, 6\}$$

$$C = \{(3, 6), (4, 5), (5, 4), (6, 3)\}$$

$$P(A) = \frac{3 \times 6}{6 \times 6} = P(B) = \frac{1}{2}$$

$$A \cap B = \{3\} \times \{1, 2, \dots, 6\}, \quad A \cap B \cap C = \{(3, 6)\}$$

$$P(A \cap B) = \frac{1 \times 6}{6^2} = \frac{1}{6}$$

$$P(A \cap B \cap C) = \frac{1}{36}$$

$$P(C) = \frac{4}{36} = \frac{1}{9}$$

$$P(A)P(B)P(C) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{9}$$

II] A_1, A_2, A_3 are independent A_1^c, A_2, A_3 are independent

A_1^c, A_2^c, A_3

A_1, A_2^c, A_3

Given $P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2)P(A_3)$ & pairwise independent

$$P(A_1^c \cap A_2 \cap A_3)$$

$$= P(A_2 \cap A_3) - P(A_1 \cap A_2 \cap A_3)$$

$$= P(A_2 \cap A_3) - P(A_1)P(A_2 \cap A_3)$$

$$= (1 - P(A_1))P(A_2 \cap A_3)$$

$$P(A_1^c \cap A_2 \cap A_3) = P(A_1^c)P(A_2)P(A_3)$$

$\Downarrow$
 A_1^c, A_2, A_3 are
pairwise independent
assume
too.

$$P(A_1 \cap A_2) = P(A_1)P(A_2)$$

$$P(A_1^c \cap A_2)$$

$$= P(A_2) - P(A_1 \cap A_2)$$

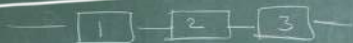
$$= P(A_2) - P(A_1)P(A_2)$$

$$= P(A_2)(1 - P(A_1))$$

$$= P(A_2)P(A_1^c)$$

Serial & Parallel Networks

Serial Network



A_i : link i fails. assume each link can fail independently

$$P_F = P(A_1 \cup A_2 \cup A_3)$$

$$= P\left(\left(A_1^c \cap A_2^c \cap A_3^c\right)^c\right) \text{ i.e., } A_1, A_2, A_3 \text{ are independent}$$

$$P(A_i) = P_i$$

$$P_F = 1 - P(A_1^c \cap A_2^c \cap A_3^c) = 1 - P(A_1^c)P(A_2^c)P(A_3^c)$$

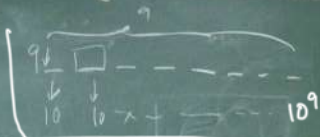
$$P_F = 1 - \prod_{i=1}^3 (1 - P_i)$$

Parallel Network



$$P_F = P(A_1 \cap A_2 \cap A_3) = P_1 P_2 P_3$$

Basics of Counting



Consider a process that contains k stages. Telephone #s that start with 9.

- ① In first stage you have n_1 possibilities
- ② For every option in first stage you have n_2 possibilities in 2nd stage.

In general for every option in first $i-1$ stages you have n_i options in the i th stage.

$$\Rightarrow \text{Total \# of possibilities} = n_1 \times n_2 \times \dots \times n_k$$

Examples of Counting

of 3 letter words that are possible: $= 26 \times 26 \times 26 = 26^3$

of distinct 3 letter words $\dots = 26 \times 25 \times 24$

of subsets of $S = \{1, 2, \dots, n\} = \underbrace{2 \times 2 \times \dots \times 2}_{n \text{ times}} = 2^n$

of function from $\{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, m\} = \underbrace{m \times m \times \dots \times m}_{n \text{ times}} = m^n$

of bijective functions

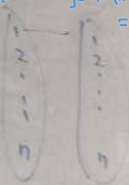
possible from $\{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, n\} = n \times (n-1) \times \dots \times 1 = n!$

of injective functions from $\{1, 2, \dots, k\} \rightarrow \{1, 2, \dots, n\} = n \times (n-1) \times \dots \times (n-k+1) = {}^n P_k$

Permutations: We have n distinct objects.

Find # of ways to pick k objects and arrange them.

$\left\{ \begin{matrix} abc \\ bac \end{matrix} \right\} \rightarrow {}_{(26)}P_3$



$${}_{(n)}P_k = \underbrace{1 \times \dots \times k}_{k \text{ objects}} = \frac{n!}{(n-k)!}$$